Gabriel **Corsi Honório**

Personal information

Electrical engineering student at ENSTA Paris and UFSCar specializing in embedded systems, robotics, and control. Experienced with STM32, ROS 2, Gazebo and real-time communication protocols (CAN, I2C, Ethernet). Seeking a 6-month internship (Apr–Oct 2026) focused on embedded development, robotics integration, autonomous systems, or industrial automation.

Contact

corsihonoriog@gmail.com

Île-de-France - FR

Links

Personal Website

LinkedIn

Github

Resources

Softwares
Gazebo, LTspice, Proteus, PlatformIO, Fusion 360, AutoCAD, Inventor, Simulink

Softskills
Strong communication, group leadership, creativity, and project management skills.

Languages

Portuguese (Native)

English (Intermediate/advanced)

French (Intermediate)

Professional Experience

Embedded Systems Intern – SNCF 05/2025 - 08/2025

- Developed a railway track inspection test bench, creating a state machine on STM32 for motor control and synchronized data acquisition. Integrated hardware and software via CAN, I2C, USART, and Ethernet, and built a Python/ROS 2 interface for real-time control and monitoring.

Academic Experience

LiDAR-Inertial Odometry Benchmark 10/2025 – ongoing

- Developing and benchmarking LiDAR-Inertial Odometry (DLIO) algorithms in ROS2 and Gazebo, with a focus on ADAS perception and localization. Building a simulated 3D LiDAR + IMU environment for data generation and evaluation using EVO.

Autonomous Vehicle Competition 11/2024 - 05/2025

- Project Leader of a 16-member team to develop a modified RC car with LIDAR-based navigation. Managed project planning, coordinated area leads, and ensured integration.

Scientific research 06/2023 - 07/2024

- Research on the sizing of a Residential Battery Energy Storage System for Revenue Maximization in Electricity Markets, considering stochastic solar photovoltaic generation.

Education

Engineering Diploma 09/2024 - 09/2026

École Nationale Supérieure de Techniques Avancées

Electrical Engineering Degree 08/2021 - Expected 09/2026

Federal University of São Carlos

Mechatronics Technician 01/2018 - 12/2020

ETEC Professor Basilides de Godoy

Skills

Programming

Python & MATLAB for data analysis, simulations, and ML.

C/C++ and FreeRTOS for embedded systems.

AMPL & Pyomo for optimization modeling. JavaScript, HTML, CSS for website development.

Embedded Systems & Robotics

Experience with ESP & STM32 for IoT and robotics. Sensor acquisition (accelerometers, encoders, Lidar), real-time data synchronization

Gazebo simulation, ROS 2 integration, CAN, I2C, USART, and Ethernet communication. System modeling and the development of classical and modern control.

3D Modeling for part and mechanism design (supports, housings, assemblies).